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ATLAS EXPERIMENT

Some Thoughts on our Background Estimate

Tobias Heintz, B2F Theory Chat, 31/10/2025

Last Week's "Next Steps"

- Understand Background Calculations better
  - Take CalRatio Data Rates instead diJet simulation (see Table 3 of JHEP06 2022 005) ← Turns out they mainly require 1 DJ at the pre-selection stage
    BUT: they require 2 jets with pT > 40 GeV … this will kill our signal sensitivity
    - ★ However, they require 2 DJs
    - ★ Can we get backwards, and disentangle the 2 jets
  - Or we could take diJet MCs, and calculate probability of displacement from shower depth ~ log(E) ← Perhaps that's the solution?
  - Think about BIB and cosmic muons, find argument why they are negligible… ← The ratio BIB/multijets is roughly 1/20 for CR
- Signal simulation ← Did not look into Signal since last week
  - Look at cutflow for beta_1 "parabola"
  - Generate higher cTau values to test linear dependence of long tail
  - Generate also larger dM10 (how large can we go?)

- Taking data rates for a background estimate could be quite challenging and shady …
- In my opinion, our main question should be: Does the 1/230 background reduction allow us to target 1DJ or not?
- In other words, is the background *increase* by requiring 1 DJ instead of 2 DJs smaller or greater than 230?
- If it is greater than 100, we don't need to care about a shady background estimate, but we have other problems.

CalRatio Data Rates

	Low-E_T selection:	Main data	BIB
Preselection:	trigger, 2 clean jets	40 743 867	2 200 854
Event cleaning:	$\sum \Delta R_{\min} > 0.5$	28 248 024	1 399 351
	low-E_T BDT > 0.05	1 288 596	44 035
	Trigger matching	1 138 961	36 266
	$-3 < t < 15$ns	1 123 239	35 245
	$\log_{10}(E_H/E_{EM}) > -1.5$ for $\text{jet}^{\text{sig}_1}$, $\text{jet}^{\text{sig}_2}$, $\text{jet}^{\text{bib}_1}$, $\text{jet}^{\text{bib}_2}$	1 038 019	33 100
	$\lvert\eta\rvert \notin [1.45, 1.55]$ for $\text{jet}^{\text{sig}_1}$, $\text{jet}^{\text{sig}_2}$	976 805	31 292
Low-E_T selection:	$H_T^{\text{miss}}/H_T < 0.6$	965 748	30 712
	$p_T(\text{jet}^{\text{sig}_{1l}}) > 80\,\text{GeV}$	315 530	10 048
	$p_T(\text{jet}^{\text{sig}_{2l}}) > 80\,\text{GeV}$	73 484	2810
	$\sum_{\text{jet}^{\text{sig}_{1l}},\text{jet}^{\text{sig}_{2l}}} \log_{10}(E_H/E_{EM}) > 2$	3375	93
	Low-E_T NN product > 0.7	307	10
Region A:		23	0
Region B:		3	0
Region C:		220	7
Region D:		61	3

Main data: "normal" CR Triggers

BIB data: CR Triggers with failed BIB removal

We have to apply the same pre-selection

- Since B2F is nearly as "subset" of CR low ET, I'm not worried about the trigger cut
- Though, requiring 2 jets with pT > 40 GeV will significantly reduce our signal sensitivity

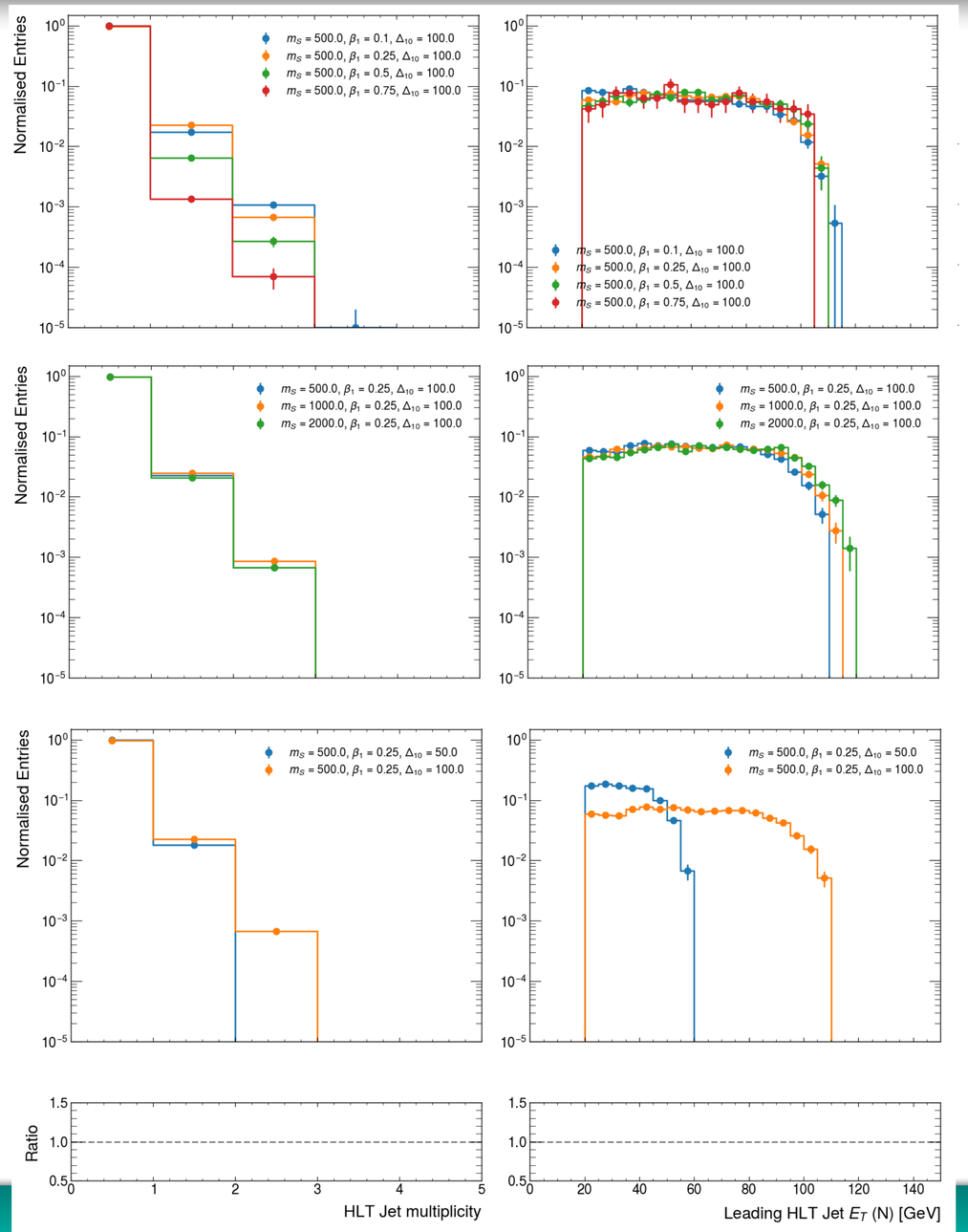
We could potentially use some of these cuts

CalRatio Workflow

- Preselection: The event has to…
  - pass at least one CR trigger
  - have a primary vertex
  - contain at least 2 clean jets: a clean jet is a jet with
    - ⋆ $p_T > 40\,\mathrm{GeV}$,
    - ⋆ $|\eta| < 2.5$, and that
    - ⋆ passes the CR jet cleaning
  - $\sum \Delta R_{\mathrm{min}}(j, \text{closest track with } p_T > 2\,\mathrm{GeV}) > 0.5$ where the sum runs over all clean jets with $p_T > 50\,\mathrm{GeV}$
- Per-jet NN taggers: classify jets as signal-, BIB-, and SM-like
  - Based on tracks, tops-clusters, and muon segments
  - Returns Signal-, BIB-, and SM-Multijet NN scores for each jet
  - Also, returns a list of the 4 jet candidates: for lowEt and highEt NN, each the highest signal- and BIB-score jet
- Per-event BDTs: discriminate Signal and Background events
  - Inputs: NN outputs, ΔR between jet candidates, H_T^{miss}/H_T, … [list of all inputs is available in Appendix D of https://cds.cern.ch/record/2863887/files/ATL-COM-PHYS-2023-557.pdf]
  - Output: BDT score for low and high ET BDTs
- Event cleaning: ensure that SM MJ is the only remaining BG
  - Low ET BDT score > 0.05
  - Trigger Matching [I did not find details about how the matching is done]
  - Jet candidate's timing $\in (-3.15)\,\mathrm{ns}$
  - Veto transition region
- Final Selection: optimise S/B discrimination
  - $H_T^{\mathrm{miss}}/H_T < 0.6$
  - NN signal score product > 0.7
  - $\sum_j \log_{10} \frac{E_H}{E_{EM}} > 2$
  - $p_T(j) > 80\,\mathrm{GeV}$
- ABCD plane: estimate SM MJ background
  - $\Delta R_{\mathrm{min}} > 1.0$
  - BDT score > 0.27

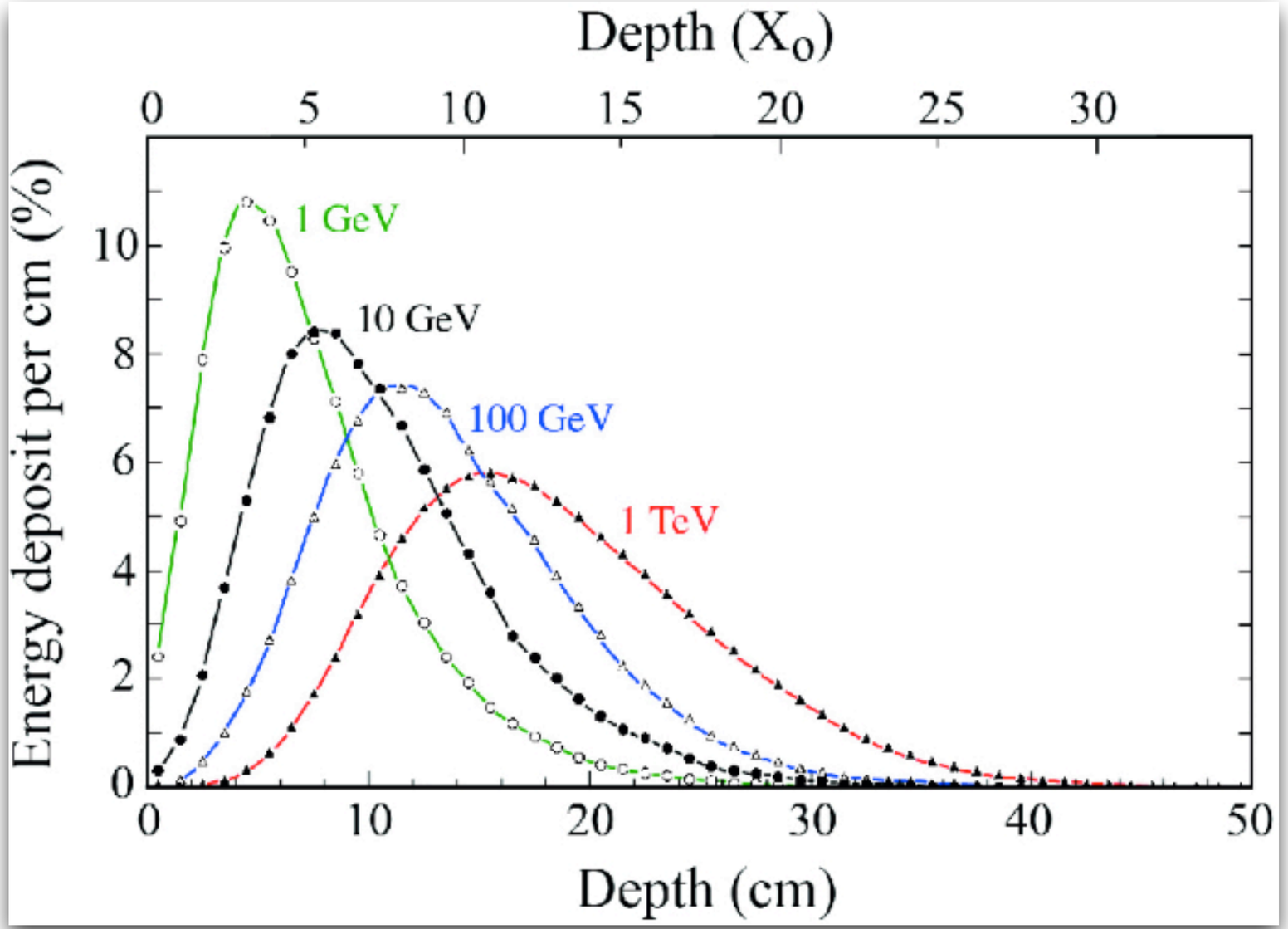
Jet Multiplicities @ MC Truth Level

- Requiring two jets at the MC Truth level, will decrease our Signal efficiency by more than ~ one order of magnitude
  - In data and MC Reco, the PU might increase the jet multiplicity… hard to judge without looking at MC Reco level



$c\tau = 20\,\mathrm{m}$

$\Delta_{21} = 7\,\mathrm{GeV}$

Backup

Shower Development